

JACL Reference Card

FUNCTION SYNTAX

<code>{name</code>	Associated function (on the preceding object or location).
<code>{+name</code>	Global function.
<code>{*name</code>	Explicitly named function (no auto-suffix).
<code>{a : b : c</code>	Multiple names for the same function.
<code>}</code>	Closes a function.
<code>execute fn<arg0<arg1...</code>	Run a function with arguments. Available inside as <code>arg[n]</code> (integer) and <code>string_arg[n]</code> (string).
<code>call fn</code>	Like execute , but does not error if the function is missing.
<code>return [value]</code>	Exit the function. Default return is true .
<code>@fn</code>	Call-count of <code>fn</code> . Bare @ is the running function's own count.

DATA DEFINITIONS

<code>object Label : Name1 Name2...</code>	
<code>plural PluralName1...</code>	
<code>has Attribute1 Attribute2...</code>	
<code>short Article Description</code>	
<code>definite DefiniteArticle</code>	
<code>long Description function</code>	
<code>parent ItemLabel</code>	
<code>mass N heavy scenery</code>	
<code>capacity N</code>	
<code>player</code>	
<code>static</code>	
<code>location Label : Name1 Name2...</code>	
<code><direction> DestLabel</code>	
<code>short Article Description</code>	
<code>has Attribute1...</code>	
<code>integer Label [Value]</code>	
<code>string Label ["Value"]</code>	
<code>constant Label Value</code>	
<code>attribute Label1 [Label2...]</code>	
<code>grammar Syntax >CoreFunction</code>	
<code>synonym OldWord NewWord</code>	
<code>filter Word</code>	
<code>parameter Name Target [min max]</code>	
<i>Grammar scope tokens: *held *here *present</i>	
<i>*inside *anywhere (single noun); **held...</i>	
<i>**anywhere (multi-noun). \$integer /</i>	
<i>\$string / \$location capture literal player input.</i>	

FUNCTION DISPATCH

<code>grammar verb >fn</code>	
1. <code>+before_fn</code>	— if it returns true, jump to step 7.
2. <code>fn_here</code>	— associated with the current location.
3. Otherwise <code>+fn</code>	— the global function.
4. If <code>+fn</code> hits override	, look for <code>fn_override_here</code> .
5. Otherwise <code>+default_fn</code> .	
6. Otherwise the rest of <code>+fn</code>	after the override runs.
7. <code>+after_fn</code> .	
<code>grammar verb *Obj1 >fn</code>	
1. <code>+before_fn</code>	(as above).
2. <code>fn_Obj1</code> .	
3. Otherwise <code>+fn</code> .	
4. On override: <code>fn_override_Obj1</code> .	
5. Otherwise <code>+default_fn</code> .	
6. Otherwise continue <code>+fn</code> .	
7. <code>+after_fn</code> .	
<code>grammar verb *Obj1 prep *Obj2 >fn</code>	
1. <code>+before_fn</code> .	
2. <code>fn_Obj2_Obj1</code> .	
3. Otherwise <code>+fn</code> .	
4. On override: <code>fn_Obj2_override_Obj1</code> .	
5. Otherwise <code>+default_fn</code> .	
6. Otherwise continue <code>+fn</code> .	
7. <code>+after_fn</code> .	

TEXT MACROS

<i>Applied as noun{macro}. Capitalise the macro name (e.g. {The}) for sentence-initial form.</i>		
<code>{list}</code>	<code>{plain}</code>	<code>{long}</code>
<code>{the}</code>	<code>{label}</code>	<code>{names}</code>
<code>{sub}</code>	<code>{obj}</code>	<code>{that}</code>
<code>{it}</code>	<code>{is}</code>	<code>{isnt}</code>
<code>{does}</code>	<code>{doesnt}</code>	<code>{s}</code>

OBJECT ELEMENTS

0 parent	8 index
1 capacity / quantity	9 status
2 mass	10 timer
3 bearing	11 counter
4 velocity	12 points
5 next	13 class
6 previous	14 x
7 child	15 y

OBJECT ATTRIBUTES

1 CLOSED	17 BURNING
2 LOCKED	18 LOCATION
3 DEAD	19 ON
4 IGNITABLE	20 DAMAGED
5 WORN	21 FEMALE
6 CONCEALING	22 POSSESSIVE
7 LUMINOUS	23 OUT_OF_REACH
8 WEARABLE	24 TOUCHED
9 CLOSABLE	25 SCORED
10 LOCKABLE	26 SITTING
11 ANIMATE	27 NPC
12 LIQUID	28 DONE
13 CONTAINER	29 GAS
14 SURFACE	30 NO_TAB
15 PLURAL	31 NOT_IMPORTANT
16 FLAMMABLE	

LOCATION ATTRIBUTES

1 VISITED	9 POLLUTED
2 DARK	10 SOLVED
3 ON_WATER	11 MID_WATER
4 UNDER_WATER	12 DARKNESS
5 WITHOUT_AIR	13 MAPPED
6 OUTDOORS	14 KNOWN
7 MID_AIR	18 LOCATION
8 TIGHT_ROPE	

POINTERS

<code>player</code>	The object that is the player.
<code>here</code>	The player's current location.
<code>noun1, noun2</code>	First and second objects in the player's command.
<code>noun3</code>	Default loop iterator.
<code>noun4</code>	Free pointer.
<code>self</code>	Object the running function is associated with.
<code>destination</code>	Where the player is moving to (set before +movement).

FLOW CONTROL

if e1 [: e2...]	True if any expression is true.
ifall e1 [: e2...]	True if all are true.
ifstring e1 [: e2...]	String version of if .
ifexecute fn	True if <i>fn</i> exists and returns true.
else / endif	Branch / close.
endall	Close every open if/ifall at once.
loop [var] ... endloop	Iterate every defined object.
select obj ... endselect	Iterate the children of an object.
repeat ... until e	Loop while e is false.
while e ... endwhile	Loop while e is true.
iterate csv ... enditerate	Walk a CSV file row by row.
update csv ... endupdate	Walk a CSV file with write-back.
break	Exit the innermost loop.
return [v]	Exit the function.

OPERATORS

= == != <>	Integer equality / inequality.
> < >= <=	Numeric comparison.
has / hasnt	Attribute is / isn't set on an object.
is / isnt	Object is / isn't in a scope (*held, *here, etc.).
grandof / !grandof	Tests deep parent relationships.
locationof / !locationof	Left is the enclosing location of right.
childof / !childof	Left is a direct child of right.

COMMON VERB FUNCTIONS

Define {verb_label} for an object-specific response, or {+default_verb} for a global override. See Appendix B of the Author's Guide for the full list of 216 verb functions.

examine	take	drop
open	close	lock
unlock	wear	remove
insert_in	insert_on	look_in
look_under	look_behind	look_through
talk_to	ask_about	ask_for
tell_about	give_to	show_to
pull	push	move
read	eat	drink_from
turn_on	turn_off	use
throw_at	break	cut
listen_to	smell	taste
climb	enter	jump
light	extinguish	burn
tie	untie	attack

COMMAND SUMMARY

Output / state

set	ensure	move	write
print	print:	style	cursor
clear	more	flush	updatestate
setstring	addstring	escapestring	padstring
length	split	points	terminate

undomove

Flow

if	ifall	else	endif
endall	ifstring	ifexecute	loop
endloop	select	endselect	repeat
until	untilall	while	endwhile
iterate	enditerate	update	endupdate
break	return	execute	call

Special / I-O

proxy	override	position	bearing
distance	dir_to	npc_to	random
asknumber	getnumber	getstring	getyesorno
savegame	restoregame	timer	sound
image	volume	stop	getenv

Web (cgijacl / fcgijacl)

hyperlink	hidden	prompt	button
option	append	append_fc	append_nt
append_lc	insert		

LIBRARY HOOKS

+intro	Runs once at game start (and re-runs while web-side pending questions are unresolved).
+eachturn	After every turn-consuming command.
eachturn	On a location, fires when the player is there.
+timer	Periodic, scheduled by the timer opcode.
+movement	Global pre-movement hook; return true to block.
movement	On a location, fires when the player tries to leave.
+title	Optional header printed before each look .
+before_look / +after_look	Wrap every look with custom content.
+no_light	Replaces output when the location has DARKNESS.
+update_status_window	Draw the status window (Glk grid & web).
+rpc	Web only: catches arbitrary ?rpc=... calls.
+game_intro	webinterface scaffold (calls +game_dedication and +game_scene).